Check for updates

Scandinavian Journal of Medicine & Science in Sports

WILEY

ORIGINAL ARTICLE OPEN ACCESS

Quadriceps Morphology 6- to 60-Months After Anterior Cruciate Ligament Tear

Justin M. Losciale[1,2,3,4,5] | Bruce Lin[6] | Marine Theret[6] | Allyson M. Schweitzer[7] | Matthew D. Fliss[7] | Thomas Scarr[8] | Xian He Yan[1,2,9] | Navin Prasad[10] | Michael A. Hunt[1,9] | Michael S. Koehle[7,10] | Fabio M. V. Rossi[6] | Cameron J. Mitchell[7] | Jackie L. Whittaker[1,2,9]

[1]Department of Physical Therapy, Faculty of Medicine, University of British Columbia, Vancouver, Canada | [2]Arthritis Research Canada, Vancouver, Canada | [3]Edwin S.H. Leong Centre for Healthy Aging, Faculty of Medicine, University of British Columbia, Vancouver, Canada | [4]Informatics, Decision-Enhancement and Analytic Sciences (IDEAS) Center of Innovation, Veterans Affairs Salt Lake City Health Care System, Salt Lake City, Utah, USA | [5]Division of Epidemiology, Department of Internal Medicine, University of Utah School of Medicine, Salt Lake City, Utah, USA | [6]School of Biomedical Engineering and Department of Medical Genetics University of British Columbia, Vancouver, British Columbia, Canada | [7]School of Kinesiology, Faculty of Education, University of British Columbia, Vancouver, Canada | [8]Allan McGavin Sports Medicine Clinic, Vancouver, Canada | [9]Centre for Aging SMART, University of British Columbia, Vancouver, Canada | [10]Division of Sport and Exercise Medicine, Department of Family Practice, University of British Columbia, Vancouver, Canada

Correspondence: Jackie L. Whittaker (jackie.whittaker@ubc.ca)

Received: 28 March 2025 | **Revised:** 10 June 2025 | **Accepted:** 4 July 2025

Funding: This study was funded by The University of British Columbia Health Innovation Funding Investment Award. F.M.V.R. receives support by CIHR foundation grant FDN-159908. J.M.L. received salary support from the Arthritis Society PhD Salary Award (21-0000000086) and the University of British Columbia Four Year Fellowship. J.L.W. is supported by a Michael Smith Foundation for Health Research a Scholar Award (SCH-2020-0403). The funder played no role in the design, conduct, or reporting of this study.

Keywords: atrophy | fiber types | fibrosis | muscle performance | quadriceps

ABSTRACT

The purpose of this study was to compare quadriceps morphology between legs and individuals with a first-time anterior cruciate ligament (ACL) tear and uninjured controls. This was an exploratory cross-sectional study. We enrolled participants who were aged 19–35 years, and 6–60 months after a first-time ACL tear as well as uninjured controls. Participants attended two sessions separated by a minimum of 7 days. Bilateral knee extensor concentric and eccentric strength and rate of torque development (RTD) were assessed at the first session, and bilateral vastus lateralis muscle biopsies were obtained at the second. Muscle fiber-type specific cross-sectional area (fCSA), fiber-type proportions, and extracellular matrix area were calculated. Between-limb and between-group differences were estimated using mixed-effects regression. 41 of 47 (87%) participants completed the study ($n = 21$ ACL, $n = 20$ control). ACL group participants were a median (range) of 23 months from injury, and 78% had undergone an ACL reconstruction. Uninjured group participants had greater knee extensor concentric strength (−39.3 Nm; 95% CI: −54.2, −20.5), eccentric strength (−30.2 Nm; 95% CI: −53.4, −11.6), and RTD (−0.16 Nm/ms; 95% CI: −0.33, −0.04). No between-group or between-limb differences in muscle fiber-type specific fCSA, fiber-type proportions, or extracellular matrix area were observed. Despite deficits in knee extensor muscle performance, no difference in vastus lateralis fiber-type specific CSA, fiber-type proportion, or extracellular matrix area was detected between individuals 6–60 months after an ACL tear and uninjured controls.

The lead author is a current employee of the Department of Veterans Affairs Medical Center in Salt Lake City, USA. This work does not represent Veteran research, was not conducted while they were employed by the Department of Veterans Affairs, nor was it conducted as part of their official duty or financially supported by the Department of Veterans Affairs. The views and opinions expressed in this work are those of the authors and do not represent the position or policy of the Department of Veterans Affairs or United States Government.

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2025 The Author(s). *Scandinavian Journal of Medicine & Science In Sports* published by John Wiley & Sons Ltd.

1 | Background

Anterior cruciate ligament (ACL) tears and subsequent reconstruction (ACLR) are linked to a persistent deficit in knee muscle performance (e.g., strength and rate of torque development; RTD) [1]. The combination of the joint injury and persistent deficits in muscle performance may predispose individuals to re-injury [2] and osteoarthritis [3].

Muscle performance is driven by muscle activation, motor neuron firing rates, tendon stiffness [4] and cross-bridge cycling, which refers to the binding and detaching of the sarcomere proteins actin and myosin to generate force [5]. Specifically, muscle strength is influenced by the amount of parallel cross-bridge cycling and RTD by the rate of cross-bridge cycling [4]. Physiologically, cross-bridge cycling is influenced by muscle cross-sectional area (CSA), muscle quality (i.e., myofibrillar density and extracellular matrix content [ECM]), and voluntary activation of muscle fibers [4]. Muscle fiber type (i.e., Type I, IIa, IIa/x) also influences cross-bridging given differences in CSA, myofibrillar density, and contractile velocity [4]. A change or deficiency in any of these elements could impact short- and long-term muscle performance.

To date, there is a large body of evidence describing neural-based deficits after ACL tear, including reduced voluntary activation [1]. In contrast, only a few studies have explored quadriceps (vastus lateralis muscle [VL]) morphology after ACL tear. These studies are restricted to within-person comparisons during the first 6 months after ACLR (knee extensor autografts) [6–8], and suggest selective Type IIa fiber atrophy and fiber type shifting toward Type IIa/x hybrid fibers in the ACLR leg [8]. Increased ECM area associated with collagen deposition has also been reported, indicating degeneration, reduced muscle quality, or developing muscle fibrosis [8–11]. Collectively, these changes could impact quadriceps muscle performance and a muscle's ability to adapt to resistance-based exercises.

Given that deficits in muscle performance can exist beyond 6 months post-ACL tear [1, 12], it would be helpful to understand if the preliminary findings of muscle fiber atrophy, fiber type shifting, and ECM area exist at later time points. It would also be valuable to understand if these morphological changes differ from uninjured controls, given documented changes to muscle performance of the uninjured leg after ACL tear [12]. This study will compare quadriceps (VL) fiber-type specific CSA, fiber-type distribution, and ECM area (fibrosis) between legs and between individuals with a first-time ACL tear and uninjured controls. This information will contribute to a better understanding of the relationship between quadriceps muscle morphology and performance after an ACL tear [13].

2 | Methods

2.1 | Study Design, Ethics and Reporting Framework

This exploratory, cross-sectional study was approved by the University of British Columbia Clinical Research Ethics Board (REB: #H22-01839) and is reported using the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement informed study reporting [14]. All participants provided written informed consent.

2.2 | Recruitment and Participants

We recruited a convenience sample of participants with a first-time ACL tear and uninjured controls between December 2022 and December 2023 from local sporting organizations, physiotherapy and orthopedic surgeon clinics, flyers, sports clubs mailing lists in Vancouver, British Columbia, and through social media (X and Facebook). To be eligible, participants had to be 19–35 years old, willing to attend two in-person data collection sessions, and have bilateral quadriceps muscle biopsies. Additional inclusion criteria for the ACL tear group were being 6–60 months post-ACL tear, and if applicable, ACLR had to occur < 13 months post-injury (to reduce time-related effects). Potential ACL participants were excluded if they had any knee surgery before the primary ACL tear (to either knee), or if applicable, ACLR involving surgical repair of > 3 ligaments (i.e., knee dislocation). Inclusion criteria for the uninjured controls were no previous knee joint injury or surgery, and participation in sport or recreational activities at least once a week over the past 36 months (self-reported). Individuals were not eligible if they had a systemic medical condition (e.g., cancer, hormone deficiencies, and bleeding disorder), were daily cigarette smokers, used anabolic steroids (lifetime) or blood thinning medication (current), had a pacemaker, or were pregnant or breastfeeding.

2.3 | Sample Size

As we did not perform an a priori sample size calculation due to a paucity of similar data (e.g., time period, between group comparison, etc.), this should be considered an exploratory, hypothesis generating study. We aimed to recruit at least 40 participants (20 per group), with a minimum of 10 males and 10 females in each study group, which exceeded previous studies [8].

2.4 | Procedures

Participants completed a study-specific questionnaire (e.g., demographics, resistance training history, sport/activity details, and injury details, Appendix S1) using a data management system (REDCap 10.9.4, Vanderbilt University, USA) before attending two in-person testing sessions, separated by a minimum of 7 days and a maximum of 2 months. At the first session, body mass and composition (bioelectrical impedance, Tanita Body Composition Analyzer, Model MC-980U+) and bilateral knee muscle strength and RTD (computerized dynamometry, Biodex System 4, Shirley, NY, USA) were tested. At the second session, bilateral VL biopsies were collected [15].

2.5 | Outcomes

2.5.1 | Knee Muscle Performance

Concentric and eccentric knee extensor and flexor strength, RTD were assessed with a computerized dynamometer (see

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Appendix S2 for a full description). In brief, participants were seated with their hips flexed to ~90°, and the trunk, hip, and working leg were secured with straps. Each task was performed bilaterally, starting on the uninjured leg (ACL group) or dominant leg (control group). A single, trained assessor with multiple years of experience in collecting muscle performance data (J.M.L.) performed all measurements.

Concentric knee extensor and flexor strength were estimated as the peak torque value (Nm) obtained across three maximal effort (concentric) isokinetic contractions at 90°/s through a knee range of motion from 0° ± 2° (full knee extension) to 100° ± 2°. Eccentric knee extensor strength was estimated as the peak torque value (Nm) obtained across three maximal effort (eccentric) isokinetic contractions at 60°/s through a knee range of motion from 30° ± 2° to 90° ± 2°.

Knee extensor RTD at 60° of knee flexion was estimated from an isometric contraction (3-s) [16]. Participants were instructed to "reach your maximum effort as fast as possible" and visual feedback of the torque curve was provided [16]. The analog torque signal was captured using an external analog-to-digital converter (National Instruments USB-6218, National Instruments Corp., Texas, USA) and recorded using a custom-made LabView Virtual Instrument (LabView: Advanced Programming Techniques, National Instruments Corp., Texas, USA). The analog torque signal was sampled at 2000 Hz [16, 17]. The recorded signal was processed using a custom-built program in MATLAB (MathWorks Inc., Natick, MA, USA; see Appendix S2 for full methods, signal processing, and signal processing test–retest reliability using 10 randomly selected trial). RTD was calculated as the linear slope of the torque-time curve from contraction onset to 100 ms (RTD_{100}), and 200 ms (RTD_{200}). The average of all valid trials was used in the analysis to improve value stability [16, 17].

2.5.2 | Vastus Lateralis Biopsy Collection and Handling

Bilateral VL muscle biopsies were obtained to assess fiber type CSA (μm^2), fiber type distribution (proportion 0–1) and ECM area (proportion 0–1) using a 5 mm Bergstrom needle modified for manual suction under local anesthesia (1% lidocaine with epinephrine) [15]. A sample of ~100–200 mg was obtained from the same location on each leg based on the distance from the superior pole of the patella. Muscle biopsies were obtained by trained and experienced sports medicine physicians (M.S.K., N.P.). All participants performed an overnight fast and refrained from any physical activity (beyond their necessary daily activities) for 48 h prior. Following collection, the samples were cleaned of fat or connective tissue and mounted in a custom aluminum mold with Optimal Cutting Temperature (Scigen Tissue-Plus) embedding medium. The tissue was then frozen in liquid nitrogen-cooled isopentane (L-methyl butane) and stored at −80°C until sectioning and immunohistochemistry (IHC) analysis.

2.6 | IHC Analysis

Frozen muscle samples were cut into 10 μm sections using a cryostat (HM 525NX, Thermo Scientific, Thermo Fischer Scientific Inc., Waltham, MA, USA) at −20°C, mounted onto positively charged slides, and air-dried at room temperature for 1 h prior to storage at −80°C. Muscle sections were stained by the same researcher (B.L.) following recommended procedures [18]. Briefly, slides were completely air-dried at room temperature for at least 30 min. A hydrophobic border was drawn around each section by using a PAP Pen. Sections were washed twice (5 min each) with phosphate-buffered saline (PBS) and then a blocking buffer was applied (2% (w/v) Bovine Serum Albumin (BSA) and 5% goat serum (v/v) in PBS) at room temperature for 60 min; then the blocking buffer was aspirated off the sections. Sections were incubated with primary antibodies against the different types of myosin heavy chain and wheat germ agglutinin (WGA) overnight at 4°C in blocking buffer (Table 1). The next day, sections were washed three times (5 min each) with PBS and incubated with secondary antibodies for 60 min at room temperature in blocking buffer (Table 1). A final wash cycle (three washes for 5 min each) with PBS followed. Finally, sections were mounted with antifade mounting medium (Fluoromount-G, SouthernBiotech, Birmingham, AL, USA), fixed with a coverslip, and stored in the dark at 4°C until imaging.

2.7 | Image Capture and Quantification

Section images and fluorophores were captured using a Nikon Eclipse Ni microscope equipped with a dual-use monochromatic and color imaging camera (Qimaging Retiga EXi for fluorescence) operated using NISElements software (Nikon, Melville, NY) by a single researcher (J.M.L). Images were taken at 10× magnification, and the stitching function was used for large images.

After images were acquired, fiber-type CSA (μm^2), fiber-type distribution (proportion 0–1) and ECM area (proportion 0–1) were estimated using a custom-built semi-automated algorithm implemented in FIJI/ImageJ (FiberSight) as described in Appendix S2.

2.8 | Confounder Selection

A directed acyclic graph (DAG) was created (www.daggity.net) to depict the cause-effect relationship between ACL tear (exposure) and muscle morphology, to transparently guide confounder selection for statistical analyses (Appendix S3) [13]. Methods of measurement are reported in Appendix S1.

2.9 | Statistical Analysis

Analyses were performed in R (Version 4.4.3, R Core Team [19]). Demographic, confounding, and outcome variables were summarized as means (SD), median (interquartile range), or counts (%), as applicable, by study group. Further stratification of muscle performance and muscle morphology data was performed by ACL injury treatment (ACLR or no ACLR), graft type (hamstring or knee extensor; inclusive of quadriceps tendon and bone-patellar tendon-bone), and sex (female or male). No formal comparisons were made due to inadequate size of each group

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | Primary and secondary antibody combinations.

Antibody (Target)	Concentration	Source	Notes
Primary			
BA-D5 (MyHC I)	1:200	DSHB[b]	IgG2b
SC-71 (MyHC IIa)	1:10	DSHB[b]	IgG1
6H-1 (MyHC IIx)	Neat	DSHB	IgM
WGA (ECM)[a]	1:50	Invitrogen W56132	—
Secondary			
Fluor 647 (MyHC I)	1:200 in blocking buffer	Invitrogen A21242	Goat anti-mouse (IgG2b)
Fluor 488 (MyHC IIa)	1:200 in blocking buffer	Invitrogen A21121	Goat anti-mouse (IgG1)
Fluor 594 (MyHC IIx)	1:200 in blocking buffer	Jackson Labs 115-585-020	Goat anti-mouse (IgM)

Abbreviations: DSHB, Developmental Studies Hybridoma Bank; ECM, extracellular matrix; MyHC, myosin heavy chain; WGA, wheat germ agglutinin.
[a]Pre conjugated primary and secondary combination with Alexa Fluor Plus 405.
[b]Growing cells were purchased, then concentrated from the medium and purified by the University of British Columbia Antibody Laboratory.

and the exploratory nature of these descriptive data. An "injured leg" was randomly assigned to control participants to enable between-group and between-leg comparisons.

To describe the magnitude of the muscle performance impairment, between-group (ACL vs. uninjured controls) and between-leg (injured vs. uninjured) differences in muscle performance measures were estimated via linear mixed-effects regression models (95% CI) adjusted for fat-free mass and current Tegner Activity Level. Random intercepts (for each participant) were used.

Muscle fiber-type specific CSA differences were estimated between groups (ACL vs. uninjured control) and between legs (injured vs. uninjured) using linear mixed-effects regression models (95% CI). These models included the fixed effects of group (ACL vs. uninjured), leg (injured vs. uninjured), group × leg interaction term, and adjustment for sex and current Tegner Activity Score. Random intercepts (for each participant) and maximum likelihood estimation were used. Maximum likelihood estimation can account for missing outcome data (21%) and is comparable to multiple imputation based on a reasonable missing at random assumption [20] since missingness of biopsy outcomes is more related to technical issues of biopsy extraction and handling.

Muscle fiber-type specific proportions (0–1) and ECM area (0–1) differences were estimated between-group (ACL vs. control) and between-leg (involved vs. uninvolved) using mixed effects beta regression models (95% CI) implemented with the 'gam' function in R using a "logit" link. Beta regression maintains the bounded nature of the outcome (between 0 and 1) and results in more accurate standard errors and confidence intervals that cannot surpass the bounds [21]. Models included the fixed effects of group (ACL vs. control), leg (injured vs. uninjured), a group × leg interaction term and adjustment for sex and Tegner Activity Scale score (for fiber type proportions) or adjusted for age and sex (ECM area). Random intercept for each participant was implemented as a random smooth [22]. Coefficients were converted from logits to interpretable beta-coefficients and presented on the percent scale (0%–100%) for clarity using the "marginaleffects" package [23]. Model assumptions (linearity, homoscedasticity, and normality of residuals) were assessed for all models and homoscedasticity was violated in most analyses. As a result, all standard errors (and 95% CIs) were estimated using bootstrap resampling with 500 replications [24].

2.10 | Sensitivity Analyses

To understand the robustness of the between-group comparisons, multiple sensitivity analyses were performed, including: (1) restricting the injured group to ACLR only participants, (2) restricting the analysis to muscle samples with > 200 fibers ($n = 17$ limbs removed), (3) reassigning the dominant leg as the injured leg for the control group, (4) changing the confounders in each analysis to weekly frequency of knee-focused resistance training and ethnicity, (5) removing extreme values (2 or more z-score) and repeating the primary analyses ($n = 3$ removed), (6) removing only uninjured control participants with larger than expected between-limb differences (over 500 μm^2) and repeating the primary analyses ($n = 7$ removed), and (7) using fiber type specific minimum ferret diameter instead of CSA as the outcome.

3 | Results

3.1 | Participants

Participant eligibility and enrollment flow chart (including attrition reasons) are provided in Appendix S4. A total of 47 participants were enrolled (ACL: 26, control: 21), and 41 completed the study (ACL: 21, control: 20; 13% attrition). More males withdrew than females ($n = 4$ vs. $n = 2$). For the ACL group, three participants provided no data after enrollment, one completed only the questionnaires, and another completed everything but the muscle biopsies. The control group participant who withdrew completed everything but the biopsies.

Participant characteristics are described in Table 2. Most participants ($n = 40$, 91%) performed knee-focused strengthening exercises at least once a week (ACL group; median 2 days/week;

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Participant characteristics.

Characteristic	ACL group ($n = 23$)	Uninjured group ($n = 21$)
Age (years)	27.0 (4.6)	24.5 (2.6)
Sex		
Female	11 (47.8)	11 (52.4)
Male	12 (52.2)	10 (47.6)
Gender		
Woman	10 (43.5)	11 (52.4)
Man	12 (52.2)	10 (47.6)
Queer	1 (4.3)	0 (0.0)
Highest education		
High school	4 (17.4)	5 (23.8)
Some college	2 (8.7)	0 (0.0)
Trade	1 (4.3)	1 (4.8)
Bachelors	12 (52.2)	11 (52.4)
Masters	4 (17.4)	4 (19.0)
Employment status[a]		
Unemployed	1	1
Self-employed	0	1
Student	8	16
Part or full-time	16	9
Income (Canadian dollars)[b]		
≤ 50 000	1 (4.3)	11 (52.4)
50 000–100 000	8 (34.8)	3 (14.3)
> 100 000	8 (34.8)	3 (14.3)
Ethnicity		
White	15 (65.2)	15 (71.4)
Non-White	8 (34.8)	6 (28.6)
Current Tegner Activity Scale[c]	6.0 (6.0, 7.0)	5.0 (4.0, 6.0)
Pre-Injury Tegner Activity Scale[c]	7.0 (6.0, 9.0)	NA
CHC use (females only)		
Yes	4 (36.4)	3 (27.3)
No	7 (63.6)	8 (72.7)
Height (cm)	171.2 (9.5)	171.1 (10.6)
Body mass (kg)	73.0 (10.5)	67.2 (11.8)
Fat free mass (kg)	59.7 (10.6)	55.3 (11.2)
Time between sessions (days)[c]	12.5 (7.3, 21.8)	16.0 (12.5, 21.0)
Injured leg		
Dominant	8 (34.8)	10 (47.6)
Non-dominant	15 (65.2)	11 (52.4)

(Continues)

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | (Continued)

Characteristic	ACL group ($n = 23$)	Uninjured group ($n = 21$)
Currently performing knee strengthening exercise		
Yes	22 (95.7)	18 (85.7)
No	1 (4.3)	3 (14.3)
Weekly knee exercise frequency (days)[c]	2.0 (2.0, 3.0)	1.0 (1.0, 2.0)
Weekly knee muscle resistance training frequency (days)[c]	3.0 (2.0, 4.0)	2.0 (1.0, 4.0)
Years of past resistance training experience (years)[c]	4.0 (1.5, 10.0)	3.0 (3.0, 5.0)

Abbreviations: ACL, anterior cruciate ligament; CHC, combined hormonal contraceptive; cm, centimeter, kg, kilograms.
[a]Able to select more than one.
[b]Missing 6 values (ACL), 4 values (uninjured).
[c]Median (interquartile limits).

control group; median 1 day/week). ACL tear treatment details are provided in Table 3. Due to freezing artifacts or inconsistent area of cross-sectionally oriented fibers, 21.4% (9/42) of the ACL group's and 20% (8/40) of the uninjured control group's muscle biopsies were unusable.

3.2 | Knee Muscle Performance

Table 4 summarizes the comparison of knee muscle performance outcomes by limb and study group. Isokinetic concentric and eccentric knee extension strength and RTD were lower in the ACL group (compared to uninjured controls), and concentric and eccentric knee extension strength was lower in the ACL tear leg (compared to uninjured leg). Appendix S4 presents descriptive data by ACL tear treatment (ACLR vs. no surgery) and graft type (hamstring vs. knee extensor). No differences were detected between ACL tear treatments.

3.3 | Vastus Lateralis Fiber CSA, Fiber Type Distribution, and ECM Area

Representative images are shown in Figure 1. The median (IQR) number of fibers per section was 313 (228). No between-group or between-leg differences were observed for VL fiber CSA, fiber type distribution, and ECM area (Table 5). These findings were consistent across sensitivity analyses, except when controls with extreme between-leg differences were removed. In that analysis, a smaller (but non-significant) injured leg type IIa and IIax fiber CSA was detected in the ACL group (Appendix S4). Descriptive data by ACL tear treatment, graft type, and sex are summarized in Appendix S4. Figures depicting data distributions are provided in Appendix S5.

TABLE 3 | ACL tear treatment details.

Characteristic	ACL group ($n = 23$)
Treatment	
ACLR	18 (78.3)
Non-operative	5 (21.7)
Graft type	
Semitendinosus tendon	13 (72.2)
BPTB or Quadriceps Tendon	5 (27.8)
Concomitant injury	
Yes	18 (78.3)
No	5 (21.7)
Time since injury (mon)[a]	23.0 (15.5, 32.3)
Time since surgery (mon)[a]	14.8 (11.5, 27.5)
Formal post-injury rehabilitation	
Yes	22 (95.7)
No	1 (4.3)
Formal post-injury rehabilitation duration (mon)[a]	8.5 (2.3, 10.0)
Formal discharge from rehabilitation	
Yes	6 (26.1)
No	12 (52.2)
Still in rehabilitation	5 (21.7)

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; BPTB, bone patellar tendon bone; mon, months.
[a]Median (interquartile limits).

4 | Discussion

This study contributes novel and vital evidence about quadriceps muscle morphology responses > 6 months (median 23-months) after an ACL tear. Minimal to no average differences in VL fiber type-specific CSA, distribution, or ECM area were observed between participants in the ACL tear and uninjured groups or between injured and uninjured legs, despite significant knee extensor muscle performance deficits. These findings suggest that the persistent knee extensor muscle performance deficits associated with ACL tears may be due to loss of myofibrillar proteins [7, 25] and/or neural-based mechanisms [1], and that restoring muscle morphology may be necessary, but insufficient by itself. This research informs future efforts to isolate the neural-based mechanism driving persistent muscle performance deficits in this patient population and therapeutic targets.

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 | Knee muscle performance outcomes by limb and study group.

Outcome	ACL group ($n = 22$)	Uninjured group ($n = 21$)	Adjusted mean difference (95% CI)[a]
Knee extension (Injured leg)			
Isokinetic concentric (Nm)	182.5 (59.8)	191.8 (58.4)	−39.3 (−54.2, −20.5)
Isokinetic concentric (Nm/kg)	2.5 (0.6)	2.8 (0.4)	—
Isokinetic eccentric (Nm)	254.5 (75.4)	252.1 (58.4)	−30.2 (−53.4, −11.6)
Isokinetic eccentric (Nm/kg)	3.5 (0.9)	3.7 (0.5)	—
RTD_{100} (Nm/ms)	0.73 (0.26)	0.83 (0.31)	−0.16 (−0.33, −0.04)
RTD_{200} (Nm/ms)	0.72 (0.22)	0.77 (0.23)	−0.12 (−0.22, −0.04)
Knee flexion (Injured leg)			
Isokinetic concentric (Nm)	86.2 (27.2)	76.0 (26.0)	0.4 (−6.4, 8.1)
Isokinetic concentric (Nm/kg)	1.2 (0.3)	1.1 (0.2)	—
Knee extension (Uninjured leg)			
Isokinetic concentric (Nm)	210.3 (61.6)	189.5 (59.0)	−9.3 (−24.9, 3.0)
Isokinetic concentric (Nm/kg)	2.9 (0.6)	2.8 (0.4)	—
Isokinetic eccentric (Nm)	292.5 (68.6)	245.6 (63.0)	14.4 (−7.9, 35.7)
Isokinetic eccentric (Nm/kg)	4.0 (0.7)	3.6 (0.5)	—
RTD_{100} (Nm/ms)	0.78 (0.41)	0.79 (0.30)	−0.06 (−0.20, 0.07)
RTD_{200} (Nm/ms)	0.80 (0.28)	0.75 (0.25)	0.00 (−0.08, 0.08)
Knee flexion (Uninjured leg)			
Isokinetic concentric (Nm)	92.3 (30.3)	76.2 (20.9)	6.2 (−0.6, 13.1)
Isokinetic concentric (Nm/kg)	1.3 (0.3)	1.1 (0.2)	—
Limb symmetry index (%)			
KE isokinetic concentric[b]	88.0 (84.0, 96.0)	99.0 (95.0, 111.0)	—
KE isokinetic eccentric[b]	88.0 (80.0, 96.0)	99.0 (96.0, 113.0)	—
KE RTD_{100}[b]	102.0 (76.0, 125.0)	100.1 (90.0, 133.0)	—
KE RTD_{200}[b]	91.0 (76.0, 103.0)	100.1 (89.0, 126.0)	—
Between-leg differences (mean, 95% CI, uninjured leg as reference)			
KE isokinetic concentric (Nm)	−27.7 (−46.5, −13.7)	2.4 (−10.2, 13.6)	−30.1 (−45.9, −14.3)
KF isokinetic concentric (Nm)	−6.0 (−14.2, 2.2)	−0.2 (−4.7, 5.2)	−5.8 (−13.5, 1.8)
KE isokinetic eccentric (Nm)	−38.0 (−61.5, −18.2)	6.6 (−8.7, 20.6)	−44.5 (−65.8, −23.3)
KE RTD_{100} (Nm/ms)	−0.06 (−0.24, 0.07)	0.04 (−0.08, 0.20)	−0.1 (−0.3, 0.1)
KE RTD_{200} (Nm/ms)	−0.09 (−0.20, 0.02)	0.02 (−0.06, 0.09)	−0.12 (−0.22, −0.01)

Note: Data are mean (standard deviation).
Abbreviations: ACL, anterior cruciate ligament; CI, confidence interval; KE, knee extensor; kg, kilogram; ms, milliseconds; Nm, Newton-meter; RTD, rate of torque development.
[a]Adjusted for current Tegner Activity Score and fat-free mass (kg), uninjured group was the reference, all 95% CIs are from bootstrapped standard errors with 500 repetitions.
[b]Median (interquartile limits).

4.1 | Quadriceps Morphology After ACL Tear

This study contributes important information that furthers our understanding of the time course of quadriceps muscle fiber type-specific changes after ACL tear and ACLR. Past studies have reported Type IIa fiber atrophy before [6–8], and at 6-months [7, 8], and 5 years post-ACLR [26]. The present results contrast with previous results at 5 years post-ACLR due to

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

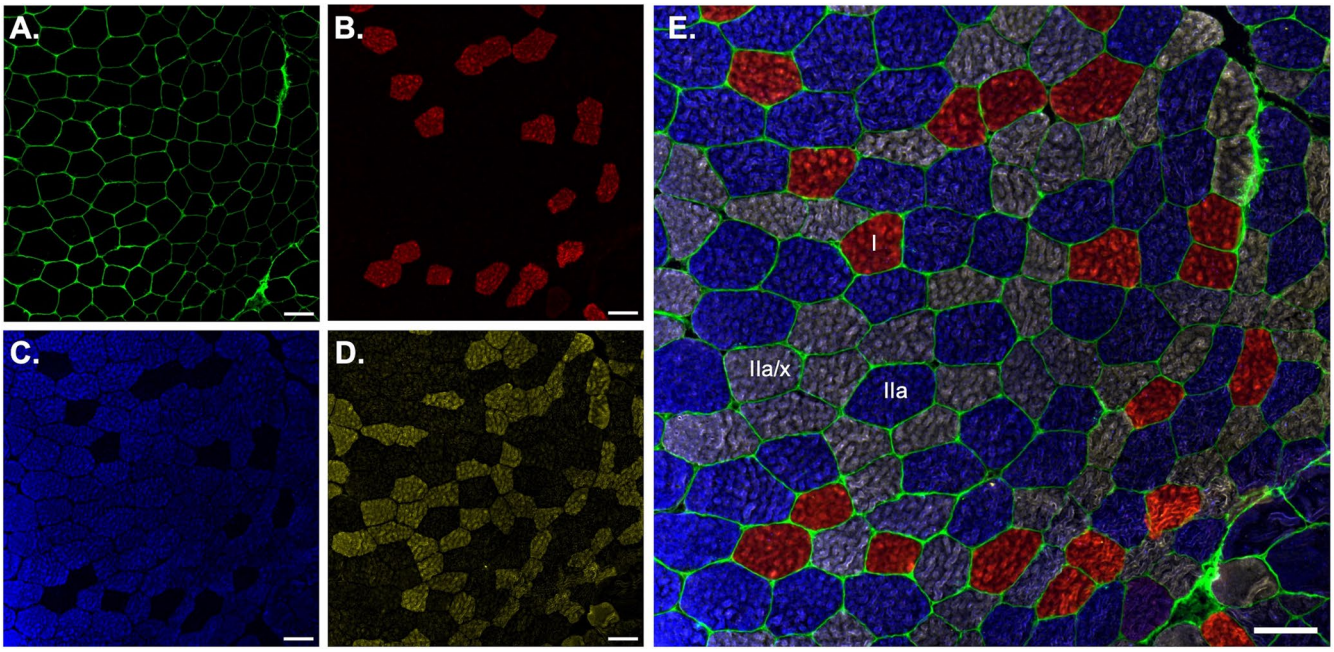

FIGURE 1 | Representative muscle biopsy image. Image from an uninjured control, scale bar is 100 μm. Panel A: wheat germ agglutinin stain for ECM area, Panel B: type I fibers, Panel C: type IIa fibers, Panel D: type IIx fibers, Panel E: composite image (Type I = red, Type IIa = blue, Type IIa/x = Gray, ECM = Green). ECM, extracellular matrix.

critical differences in operative and postoperative management between the current and previous study [26] which involved open ACLR and had a 7-week cast immobilization after surgery [26]. Both treatments are not currently standard of care [27, 28] and likely contributed to a compounding effect on muscle atrophy and quality [29, 30]. Collectively, the present findings suggest VL fCSA atrophy may resolve over time with consistent knee focused resistance-based exercise.

Specifically, ACL group participants in the current study self-reported they were a median 23 months post-ACL tear, had engaged in a median of 8.5 months of formal rehabilitation, and were currently doing knee muscle focused resistance training a median of 2 days/week (which is consistent with the American College of Sports Medicine (ACSM) [31] guidelines). This length of rehabilitation and weekly resistance training is likely not representative of practice patterns across Canada, the United States (57% of surveyed physical therapists report ACL tear rehabilitation < 5-months) [32] or other parts of the world, and may reflect participants education level (70% had a Bachelor's degree or higher) and other demographic features that impact exercise behaviors (e.g., socio-economic status) [6–9, 11, 33, 34]. The notion that rehabilitation length and continued resistance training beyond rehabilitation may be important for restoring quadriceps muscles CSA is also supported by the null finding of a recent randomized trial [33] where 4–5 months of 3 days/week of ACSM resistance-based rehabilitation from 1 month pre-ACLR to 4 months post-ACLR resulted in no increase in VL fCSA (across fiber types). Collectively, these findings suggest that rehabilitation and continued knee muscle focused resistance training beyond supervised rehabilitation may be needed to overcome muscle atrophy after ACL tear. The consistency of the findings across our sensitivity analyses further supports the robustness of this hypothesis.

Another possible explanation of no between-group differences in VL morphology could also be the low prevalence of participants with knee extensor grafts (28%). Past studies have focused almost entirely on individuals with bone-patellar tendon-bone-ACL grafts [7, 8, 33], a procedure that removes one-third of the patellar tendon, potentially compounding extensor weakness [35]. It is possible that graft type may mediate the effects of ACLR on quadriceps fCSA [36] which is supported by our DAG and descriptive data for VL Type IIa fiber fCSA (see Appendix S4). With that said, no formal analysis was performed to compare graft types due to no a priori plan for these analyses and low power. While the exact mechanism of action is unknown, there could be similarities to the effects seen in people with patellar tendinopathy [37] or differences in rehabilitation protocols that are based on graft-type [38].

Our findings of no VL fiber-type proportion differences between injured and uninjured legs are consistent with two recent studies [6, 7]. Collectively, these data suggest that if any fiber-type shifting does occur, it is at the individual level and likely a function of activity-dependent mechanisms such as leg unloading/immobilization and/or disuse [29], and not a result of an ACL tear unique mechanism.

In contrast to past research, we did not find evidence of an expanded ECM area in the ACL tear leg compared to both the uninjured leg and uninjured controls. A finding that was robust across our sensitivity analyses, with 95% CIs indicating differences in both directions are possible. This finding could be explained by our larger sample size ($n = 23$ vs. $n = 10$) [8, 9, 11], older participant age, and frequency of knee-focused exercise. Given the exploratory nature and approach of this and past research, we should not jump to the conclusion that ECM expansion, increased collagen accumulation, changes to the dominant collagen type, or collagen disorganization do not occur after ACL tears [10]. To definitively

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 | Vastus lateralis morphology outcomes by limb and study group.

	ACL group ($n = 21$)[a]	Uninjured group ($n = 20$)[a]	Adjusted mean difference (95% CI)[b]
Injured leg			
Type I fiber CSA (μm^2)	7396 (1153)	7099 (1146)	194 (−618, 943)
Type IIa fiber CSA (μm^2)	8011 (2156)	7788 (1711)	40 (−1431, 736)
Type IIa/x fiber CSA (μm^2)	7137 (1982)	6753 (1341)	335 (−705, 1113)
Type I fiber proportion (%)	43.5 (14.0)	43.3 (15.2)	3.3 (−5.8, 10.6)
Type IIa fiber proportion (%)	41.9 (11.2)	41.9 (15.6)	−3.6 (−10.5, 5.9)
Type IIa/x fiber proportion (%)	14.6 (6.4)	14.8 (8.7)	1.0 (−3.4, 6.2)
ECM area/total CSA (%)	13.4 (2.4)	12.6 (2.7)	0.8 (−1.3, 3.0)[c]
Uninjured leg			
Type I fiber CSA (μm^2)	7425 (1670)	7241 (1679)	−76 (−936, 758)
Type IIa fiber CSA (μm^2)	8458 (2059)	8129 (2097)	−137 (−1431, 736)
Type IIa/x fiber CSA (μm^2)	7313 (1595)	7344 (1766)	−446 (−1296, 480)
Type I fiber proportion (%)	46.7 (15.4)	44.6 (16.2)	3.8 (−4.3, 13.2)
Type IIa fiber proportion (%)	38.4 (13.5)	41.0 (13.1)	−4.5 (−12.3, 3.2)
Type IIa/x fiber proportion (%)	14.9 (7.9)	14.4 (6.7)	0.6 (−4.0, 6.2)
ECM area/total CSA (%)	13.3 (3.0)	11.6 (3.2)	1.4 (−0.5, 3.7)[c]
Within-person between-leg differences (mean, 95% CI, uninjured leg is the reference)			
Type I fiber CSA (μm^2)	82 (−1012, 741)	−188 (−1260, 796)	261 (−872, 1393)
Type IIa fiber CSA (μm^2)	−223 (−1368, 749)	−401 (−1249, 1409)	177 (−1217, 1571)
Type IIa/x fiber CSA (μm^2)	31 (−1541, 821)	−751 (−1690, 135)	770 (−322, 1862)
Type I fiber proportion (%)	−2.7 (−9.7, 0.5)	−2.2 (−1.1, 10.8)	−0.5 (−9.9, 8.8)
Type IIa fiber proportion (%)	3.8 (−4.7, 10.1)	2.8 (−7.3, 12.2)	0.9 (−8.1, 9.9)
Type IIa/x fiber proportion (%)	−0.4 (−4.0, 3.4)	−0.8 (−6.7, 5.4)	0.4 (−4.9, 5.7)
ECM area/total CSA (%)[c]	0.2 (−1.2, 1.7)	0.8 (−1.4, 4.7)	−0.6 (−2.8, 1.6)

Abbreviations: μm, micrometer; CI, confidence interval; CSA, cross sectional area; ECM, extracellular matrix.
[a]Four biopsies on the injured and uninjured leg were not usable (uninjured controls) and four biopsies (injured) and five biopsies (uninjured) were not usable (ACL group) data are mean (standard deviation).
[b]Adjusted for sex (male vs. female) and current Tegner Activity Scale score, uninjured group was the reference, confidence intervals determined via bootstrapped resampling with 500 iterations.
[c]Adjusted for age and sex (male vs. female), uninjured group was the reference.

answer those questions, further research with an appropriate sample size (leveraging the data reported in this study), additional experimental procedures (e.g., specific collagen-type stains) and adequate control of confounding factors is necessary.

4.2 | Limitations

This exploratory study was designed to generate hypotheses to inform future research aimed at unraveling the mechanisms underlying persistent muscle performance deficits after an ACL tear. There was no a priori sample size calculation, and the results should be interpreted cautiously. A future, fully powered study informed by these findings is necessary to more definitively answer the research questions explored here. The inclusion of both individuals with and without ACLR created heterogeneity in the sample but adds to the external validity (generalizability) of the results as non-operative management is becoming more common. Our sensitivity analyses suggest this decision did not impact the present results and adds valuable preliminary data for this comparison.

Consistent with past studies, muscle biopsies were taken from one quadriceps muscle (VL) at one point in time and may not be generalizable to the other quadriceps muscles, measures of whole muscle CSA or volume, or be free from measurement error [39]. We have transparently reported participant attrition and missing biopsy data (which we considered in our statistical analysis) to

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

facilitate realistic future a priori sample sizes. To address the potential for selection bias due to fewer competitive athletes in the uninjured group, the Tegner Activity Score was accounted for in statistical models. We did not purposively recruit to achieve equal numbers of participants with knee extensor and knee flexor ACL grafts as past research has detected morphology differences before ACLR [6, 8]. This led to an imbalance of graft types which prevented us from a more detailed exploration of why muscle performance and morphology appeared to differ by graft type (hamstring or knee extensor graft). Lastly, we used WGA to stain for the ECM area. This stain is a suitable alternative to the common picrosirius red stain [40] but is sensitive to all glycoproteins (not specific to interstitial collagen isoforms), which may be necessary to understand ECM collagen organization.

4.3 | Research Implications

The data from this study can be used to support sample size calculations for between-group and between-leg comparisons in muscle CSA, fiber type-specific proportions, and ECM area in individuals post-ACL tear. The use of uninjured controls presented a challenge related to selection bias, where uninjured competitive athletes were less likely to participate, which may be due to the invasive nature of the muscle biopsies. A potential solution is to collect important demographic information (e.g., age, sex, anthropometrics, exercise habits, sport type and competition level) on all individuals who are eligible (both those that do and don't enroll) so the sample can be weighted to account for selection bias [41]. Based on preliminary descriptive data, exploration of graft type-specific muscle morphology changes is warranted. The mechanism in which these graft-type differences occur should also be explored and could be related to pathophysiology (e.g., pain, tendon structure) [42] or a behavior (e.g., longer wait time to start isolated, resisted quadriceps strengthening compared to hamstring grafts) [38]. Future work should also consider simultaneous microscopic (i.e., biopsies for fCSA) and macroscopic methods (e.g., imaging) due to the normal variability in muscle biopsies based on the location of the VL [39, 43], potential non-uniform muscle atrophy after ACL tears [36] and to provide a holistic view of muscle size, as well as understand the correlation between the two different measures.

4.4 | Perspectives

This study suggests that VL fiber-type specific CSA, fiber type proportion, or ECM area of individuals a median of 23 months post-ACL tear may not differ between legs or when compared to uninjured controls, despite the presence of knee extensor muscle performance deficits. These findings deepen our understanding of quadriceps muscle fiber type and ECM area after ACL tear and ACLR and provide future topics for inquiry related to resistance training beyond formal rehabilitation and the influence of ACLR graft type.

Author Contributions

Conceptualization: J.M.L., J.L.W., M.A.H., C.J.M., M.T., F.M.V.R. (lead), all other authors (support). Methodology: J.M.L., J.L.W., C.J.M., F.M.V.R., M.T., M.S.K., (lead), A.M.S., M.D.F., B.L. Software: F.M.V.R., M.T. (lead), B.L. (support). Validation: J.M.L. (lead). Formal analysis: J.M.L. (lead). Investigation: J.M.L., B.L. (lead), A.M.S., M.D.F., X.H.Y., C.J.M., M.T., M.S.K., T.S. (support). Resources: C.J.M., FR, J.L.W. (lead), M.T., T.S. (support). Data curation: J.M.L. (lead), B.L., A.M.S., M.D.F., X.H.Y., M.S.K., N.P., T.S., C.J.M. (support). Writing – original draft: J.M.L. (lead), J.L.W., C.J.M., M.A.H. (support). Writing – reviewing and editing: all authors. Visualization: J.M.L. (lead), B.L. (support). Supervision: J.L.W. (lead), M.A.H., F.M.V.R., M.T., C.J.M. (support). Project administration: J.M.L. (lead), J.L.W. (support). Funding acquisition: J.M.L. and J.L.W. (lead), M.S.K., M.T., B.L., A.M.S., C.J.M., F.M.V.R. (support).

Disclosure

Patient and Public Involvement: A physical therapist with 4 years' of clinical experience was involved in the design, conduct, reporting, and dissemination of this current study.

Ethics Statement

This study was approved by the University of British Columbia Clinical Research Ethics Board, Vancouver, Canada (REB#H22-01839).

Consent

All participants provided written informed consent.

Conflicts of Interest

The authors declare no conflicts of interest. J.M.L. is an Associate Editor with the Journal of Orthopedic & Sports Physical Therapy Methods. M.S.K. is an editor with the Scandinavian Journal of Medicine & Science in Sports. J.L.W. is a Senior Editor with the Journal of Orthopedic and Sport Physical Therapy, Senior Associate Editor of the British Journal of Sports Medicine, Arthritis Society (Canada) Integrated Scientific and Medical Advisory Committee member, and Data Safety and Monitoring Committee member for the PIKASO trial. M.A.H. is a Data Safety and Monitoring Board Committee member for the Surgical Timing and Rehabilitation (STaR) for Multiple Ligament Knee Injuries (MLKI): A Multicenter Integrated Clinical Trial.

Data Availability Statement

De-identified data can be made available upon reasonable request to the corresponding author. Please email the corresponding author with a written proposal outlining the specific research aims, analysis plan, and why this specific data is needed. A formal data sharing agreement between institutions will be required.

References

1. B. Tayfur, C. Charuphongsa, D. Morrissey, and S. C. Miller, "Neuromuscular Function of the Knee Joint Following Knee Injuries: Does It Ever Get Back to Normal? A Systematic Review With Meta-Analyses," *Sports Medicine* 51 (2021): 321–338.

2. H. Grindem, L. Snyder-Mackler, H. Moksnes, L. Engebretsen, and M. A. Risberg, "Simple Decision Rules Can Reduce Reinjury Risk by 84% After ACL Reconstruction: The Delaware-Oslo ACL Cohort Study," *British Journal of Sports Medicine* 50 (2016): 804–808.

3. J. L. Whittaker and E. M. Roos, "A Pragmatic Approach to Prevent Post-Traumatic Osteoarthritis After Sport or Exercise-Related Joint Injury," *Best Practice and Research: Clinical Rheumatology* 33 (2019): 158–171.

4. P. Cormie, M. R. McGuigan, and R. U. Newton, "Developing Maximal Neuromuscular Power: Part 1 – Biological Basis of Maximal Power Production," *Sports Medicine* 41 (2011): 17–38.

5. K. Mukund and S. Subramaniam, "Skeletal Muscle: A Review of Molecular Structure and Function, in Health and Disease," *Wiley*

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Interdisciplinary Reviews. Systems Biology and Medicine 12 (2020): e1462.

6. M. C. Graham, K. L. Thompson, G. S. Hawk, C. S. Fry, and B. Noehren, "Muscle Fiber Cross-Sectional Area Is Associated With Quadriceps Strength and Rate of Torque Development After ACL Injury," *Journal of Strength and Conditioning Research* 38 (2024): e273–e279.

7. C. M. Latham, P. J. Balawender, N. T. Thomas, et al., "Overexpression of Manganese Superoxide Dismutase Mitigates ACL Injury-Induced Muscle Atrophy, Weakness and Oxidative Damage," *Free Radical Biology & Medicine* 212 (2024): 191–198.

8. B. Noehren, A. Andersen, P. Hardy, et al., "Cellular and Morphological Alterations in the Vastus Lateralis Muscle as the Result of ACL Injury and Reconstruction," *Journal of Bone and Joint Surgery. American Volume* 98 (2016): 1541–1547.

9. C. S. Fry, D. L. Johnson, M. L. Ireland, and B. Noehren, "ACL Injury Reduces Satellite Cell Abundance and Promotes Fibrogenic Cell Expansion Within Skeletal Muscle," *Journal of Orthopaedic Research* 35 (2017): 1876–1885.

10. B. Noehren, P. A. Hardy, A. Andersen, et al., "T1rho Imaging as a Non-Invasive Assessment of Collagen Remodeling and Organization in Human Skeletal Muscle After Ligamentous Injury," *Journal of Physiology* 599 (2021): 5229–5242.

11. B. D. Peck, C. R. Brightwell, D. L. Johnson, M. L. Ireland, B. Noehren, and C. S. Fry, "Anterior Cruciate Ligament Tear Promotes Skeletal Muscle Myostatin Expression, Fibrogenic Cell Expansion, and a Decline in Muscle Quality," *American Journal of Sports Medicine* 47 (2019): 1385–1395.

12. J. M. Losciale, C. Y. Le, N. Jansen, et al., "Strength Setbacks: The Impact of Youth Sport-Related Knee Joint Injuries on Thigh Muscle Strength. A 24-Month Prospective Cohort Study," *Journal of Orthopaedic & Sports Physical Therapy* 55 (2025): 137–147.

13. J. M. Losciale, G. S. Bullock, G. S. Collins, et al., "Description, Prediction, and Causation in Sport and Exercise Medicine Research: Resolving the Confusion to Improve Research Quality and Patient Outcomes," *Journal of Orthopaedic and Sports Physical Therapy* 53 (2023): 381–387.

14. J. P. Vandenbroucke, E. von Elm, D. G. Altman, et al., "Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): Explanation and Elaboration," *PLoS Medicine* 4 (2007): e297.

15. M. A. Tarnopolsky, E. Pearce, K. Smith, and B. Lach, "Suction-Modified Bergstrom Muscle Biopsy Technique: Experience With 13,500 Procedures," *Muscle & Nerve* 43 (2011): 717–725.

16. N. A. Maffiuletti, P. Aagaard, A. J. Blazevich, J. Folland, N. Tillin, and J. Duchateau, "Rate of Force Development: Physiological and Methodological Considerations," *European Journal of Applied Physiology* 116 (2016): 1091–1116.

17. T. Grindstaff, M. Palimenio, M. Frano, D. Anderson, J. J. Bagwell, and D. Katsavelis, "Optimizing Between-Session Reliability for Quadriceps Peak Torque and Rate of Torque Development Measures," *Journal of Strength and Conditioning Research* 33 (2018): 1840–1847.

18. K. A. Murach, C. M. Dungan, K. Kosmac, et al., "Fiber Typing Human Skeletal Muscle With Fluorescent Immunohistochemistry," *Journal of Applied Physiology (1985)* 127 (2019): 1632–1639.

19. R Core Team, *R: A Language and Environment for Statistical Computing* (R Foundation for Statistical Computing, 2021), https://www.R-project.org/.

20. J. Twisk, M. de Boer, W. de Vente, and M. Heymans, "Multiple Imputation of Missing Values Was Not Necessary Before Performing a Longitudinal Mixed-Model Analysis," *Journal of Clinical Epidemiology* 66 (2013): 1022–1028.

21. F. Cribari-Neto and A. Zeileis, "Beta Regression in R," *Journal of Statistical Software* 34 (2010): 1–24.

22. E. J. Pedersen, D. L. Miller, G. L. Simpson, and N. Ross, "Hierarchical Generalized Additive Models in Ecology: An Introduction With mgcv," *PeerJ* 7 (2019): e6876.

23. V. Arel-Bundock, N. Greifer, and A. Heiss, "How to Interpret Statistical Models Using Marginaleffects for R and Python," *Journal of Statistical Software* 111 (2024): 1–32.

24. M. A. Mansournia, M. Nazemipour, A. I. Naimi, G. S. Collins, and M. J. Campbell, "Reflection on Modern Methods: Demystifying Robust Standard Errors for Epidemiologists," *International Journal of Epidemiology* 50 (2021): 346–351.

25. T. W. Tourville, T. B. Voigt, R. H. Choquette, et al., "Skeletal Muscle Cellular Contractile Dysfunction After Anterior Cruciate Ligament Reconstruction Contributes to Quadriceps Weakness at 6-Month Follow-Up," *Journal of Orthopaedic Research* 40 (2022): 727–737.

26. M. Fluck, C. Viecelli, A. M. Bapst, et al., "Knee Extensors Muscle Plasticity Over a 5-Years Rehabilitation Process After Open Knee Surgery," *Frontiers in Physiology* 9 (2018): 1343.

27. R. Kotsifaki, V. Korakakis, E. King, et al., "Aspetar Clinical Practice Guideline on Rehabilitation After Anterior Cruciate Ligament Reconstruction," *British Journal of Sports Medicine* 57 (2023): 500–514.

28. D. S. Logerstedt, D. Scalzitti, M. A. Risberg, et al., "Knee Stability and Movement Coordination Impairments: Knee Ligament Sprain Revision 2017," *Journal of Orthopaedic and Sports Physical Therapy* 47 (2017): A1–A47.

29. E. A. Nunes, T. Stokes, J. McKendry, B. S. Currier, and S. M. Phillips, "Disuse-Induced Skeletal Muscle Atrophy in Disease and Nondisease States in Humans: Mechanisms, Prevention, and Recovery Strategies," *American Journal of Physiology. Cell Physiology* 322 (2022): C1068–C1084.

30. H. Vikne, V. Strom, A. H. Pripp, and T. Gjovaag, "Human Skeletal Muscle Fiber Type Percentage and Area After Reduced Muscle Use: A Systematic Review and Meta-Analysis," *Scandinavian Journal of Medicine & Science in Sports* 30 (2020): 1298–1317.

31. Medicine ACoS, "American College of Sports Medicine Position Stand. Progression Models in Resistance Training for Healthy Adults," *Medicine and Science in Sports and Exercise* 41 (2009): 687–708.

32. E. M. Greenberg, E. T. Greenberg, J. Albaugh, E. Storey, and T. J. Ganley, "Rehabilitation Practice Patterns Following Anterior Cruciate Ligament Reconstruction: A Survey of Physical Therapists," *Journal of Orthopaedic and Sports Physical Therapy* 48 (2018): 801–811.

33. L. N. Erickson, M. K. Owen, K. R. Casadonte, et al., "The Efficacy of Blood Flow Restriction Training to Improve Quadriceps Muscle Function After ACL Reconstruction," *Medicine and Science in Sports and Exercise* 57 (2024): 227–237.

34. J. T. Kari, J. Viinikainen, P. Bockerman, et al., "Education Leads to a More Physically Active Lifestyle: Evidence Based on Mendelian Randomization," *Scandinavian Journal of Medicine & Science in Sports* 30 (2020): 1194–1204.

35. K. S. Sinding, T. G. Nielsen, L. G. Hvid, M. Lind, and U. Dalgas, "Effects of Autograft Types on Muscle Strength and Functional Capacity in Patients Having Anterior Cruciate Ligament Reconstruction: A Randomized Controlled Trial," *Sports Medicine* 50 (2020): 1393–1403.

36. N. Ito, J. A. Martin, M. R. Joachim, et al., "Quantifying Muscle Volume Deficits Among 38 Lower Extremity Muscles in Collegiate Football Athletes After Anterior Cruciate Ligament Reconstruction," *American Journal of Sports Medicine* 53 (2025): 809–816.

37. P. Malliaras, J. Cook, C. Purdam, and E. Rio, "Patellar Tendinopathy: Clinical Diagnosis, Load Management, and Advice for Challenging Case Presentations," *Journal of Orthopaedic and Sports Physical Therapy* 45 (2015): 887–898.

38. B. Solie, J. Monson, and C. Larson, "Graft-Specific Surgical and Rehabilitation Considerations for Anterior Cruciate Ligament

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Reconstruction With the Quadriceps Tendon Autograft," *International Journal of Sports Physical Therapy* 18 (2023): 493–512.

39. F. Van de Casteele, R. Van Thienen, O. Horwath, et al., "Does One Biopsy Cut It? Revisiting Human Muscle Fiber Type Composition Variability Using Repeated Biopsies in the Vastus Lateralis and Gastrocnemius Medialis," *Journal of Applied Physiology* 2024, no. 137 (1985): 1341–1353.

40. B. Emde, A. Heinen, A. Godecke, and K. Bottermann, "Wheat Germ Agglutinin Staining as a Suitable Method for Detection and Quantification of Fibrosis in Cardiac Tissue After Myocardial Infarction," *European Journal of Histochemistry* 58 (2014): 2448.

41. M. A. Metten, N. Costet, L. Multigner, J. F. Viel, and G. Chauvet, "Inverse Probability Weighting to Handle Attrition in Cohort Studies: Some Guidance and a Call for Caution," *BMC Medical Research Methodology* 22 (2022): 45.

42. N. Ito, H. B. Sigurethsson, D. H. Cortes, L. Snyder-Mackler, and K. G. Silbernagel, "Regional Healing Trajectory of the Patellar Tendon After Bone-Patellar Tendon-Bone Autograft Harvest for Anterior Cruciate Ligament Reconstruction," *Journal of Orthopaedic Research* 42 (2024): 1399–1408.

43. O. Horwath, H. Envall, J. Roja, et al., "Variability in Vastus Lateralis Fiber Type Distribution, Fiber Size, and Myonuclear Content Along and Between the Legs," *Journal of Applied Physiology* 2021, no. 131 (1985): 158–173.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

16000838, 2025, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70109 by Ohio State University University Libraries, Wiley Online Library on [21/07/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License